

NG Feedgas Production Tracker

Project Audit Report

Auditor: Claude Date: 2026-05-28 Scope: code review + data accuracy + runtime

1. Executive summary

The project does what its design claims: it scrapes 9 public pipeline EBB portals, normalizes to a canonical FlowRecord, writes to SQLite, runs validators, and renders a Plotly Dash dashboard. The architecture is sound and the design choices (config-driven YAML, per-scraper isolation, idempotent upserts, scraper-agnostic downstream code) are good. It is currently producing useful, broadly directional numbers, and the dashboard is working.

That said, the data is not accurate at the level a market analyst would expect, and there are several active runtime bugs that distort the numbers right now. The most important findings, in order of severity:

- **[CRITICAL] TETCO scraper silently captures the wrong gas day.** Every pull falls back to "TIMELY_2026-05-27_1504" even when 2026-05-28 is requested — Sabine Pass TETCO, Freeport Stratton Ridge, and Cameron Sempra (~670 MMcf/d combined) are stored under the wrong date.
- **[CRITICAL] The Part-5 prompt is internally inconsistent.** Per-terminal lines sum to 8,758 MMcf/d but "TOTAL U.S." prints 10,478 — a hidden 1,720 MMcf/d of Mexico + Canada flows is mixed into the total without being shown in the breakdown.
- **[CRITICAL] SQLite "database is locked" races between intraday-fast (5 min) and intraday-slow (15 min) tasks.** Two completed task runs failed with exit=1 in the last 24h. No WAL mode and no busy_timeout configured.
- **[HIGH] Coverage is ~50% of true U.S. LNG feedgas.** Scraped LNG sum 8,758 MMcf/d vs EIA-reported ~17.6 Bcf/d (April 2026). Calcasieu Pass, Golden Pass, Cheniere CCPL (Corpus Christi), Cameron CIP, and DTI feeders to Cove Point are all entirely missing.
- **[HIGH] AIS bounding boxes for Cameron LNG and Calcasieu Pass overlap.** Vessels physically at Calcasieu Pass berth fall into Cameron's box first (dict-iteration order), so Calcasieu activity is undercounted and Cameron is over-counted.
- **[HIGH] Plaquemines nameplate is stale (2,000 MMcf/d vs ~3,600 actual).** Validator flags every Plaquemines pull as ">10% over nameplate" — pure noise.

Cross-validated against EIA, RBN, NGI, S&P Global, and Venture Global filings (sources in section 6).

2. Data accuracy — cross-checks vs. external sources

Most recent gas day in the DB: 2026-05-28 (evening cycle). The scrape totals were compared against EIA monthly LNG export totals and industry reporting (RBN Energy, Natural Gas Intelligence, S&P Global Commodity Insights, Pipeline & Gas Journal).

2.1 U.S. LNG feedgas — scraped vs. truth

Terminal	Scraped	Industry / EIA	% captured	Notes
Sabine Pass	2,567	~3,400	~75%	Creole Trail (Cheniere-private) missing; ET-iPost captures partial Trunkline only.
Corpus Christi	623	~2,500	~25%	CCPL (Cheniere-private) is the dominant feed. Only NGPL/Sinton is captured.
Freeport LNG	1,283	~2,200-2,500	~55%	TX intrastate feeders (Gulf South) not captured.
Cameron LNG	997	~1,400-1,700	~60%	Cameron Interstate (CIP) private. CGT meter for "Cameron LNG" returns 0 — verify row match.
Cove Point	676	~700	~96%	Transco Pleasant Valley captures the dominant feed; DTI is small.
Elba Island	120	~150	~80%	Captured via EEC.
Calcasieu Pass	0	~1,300-1,600	0%	TransCameron Pipeline shipper-private. No public EBB.
Plaquemines	2,491	~4,000	~62%	Private feeders missing. Nameplate in YAML (2,000) is stale — Phase 2 design is ~3,600.
Golden Pass	0	~700-2,500	0%	Postings on gasnom.com (ColdFusion). No scraper written.
LNG TOTAL	8,758	~17,600	~50%	Useful as a lower-bound signal, not a true U.S. LNG feedgas figure.

All MMcf/d. EIA STEO May 2026 reports 17.6 Bcf/d for April 2026 and 18.1 Bcf/d for March 2026 — the scraped totals capture roughly half of that.

2.2 Cross-border points — accuracy is much better

Border-crossing meters are single-pipe-single-meter (the project's "high confidence" tier). These match independent industry data within reasonable bounds:

- Canada — Sumas (Northwest): scraped 1,012 MMcf/d; industry typical 1.0-1.1 Bcf/d. Excellent agreement.
- Canada — Niagara (TGP delivery): scraped 578 MMcf/d (Niagara River 560 + Chippawa-side 18). YAML notes verified 540 on 2026-05-26; plausible.
- Canada — Waddington (Iroquois): scraped 124 receipt + 7.5 delivery (Brookfield). Within historical envelope (capacity up to 1.2 Bcf/d).
- Mexico — North Baja (EPNG): scraped 537 MMcf/d. Industry typical 600-700 MMcf/d. Plausible.
- Mexico — Samalayuca (EPNG): scraped 209 MMcf/d. Reasonable given 800 MMcf/d nameplate.

Texas-intrastate Mexico crossings (Trans-Pecos, Comanche Trail, NET Mexico, Valley Crossing) are not captured at all — they have no public OAC. The YAML acknowledges this and points to CENAGAS PDFs as the only public daily source. A CENAGAS PDF for January 2026 has been downloaded to data/cenagas/ but is not parsed.

3. Active runtime bugs (in priority order)

3.1 TETCO scraper picks the wrong gas day — every pull

[CRITICAL] src/ng_feedgas/scrappers/enbridge.py, lines 91-110.

CYCLE_TO_TETCO_PREFIX maps cycle="evening" to "LATEC". The portal no longer publishes LATEC_ entries on the current schedule — only TIMELY_ snapshots are present. The fallback in lines 107-109 then selects opts[0][0], which is always the most-recent TIMELY snapshot regardless of the requested gas day. The intraday-fast log shows this on every 5-minute run:

```
WARNING TETCO: no LATEC option for gas_day 2026-05-28; using first available
INFO TETCO: selected cycle option TIMELY_2026-05-27_1504
```

Consequence: Sabine Pass TETCO (~246), Freeport Stratton Ridge (~136), and Cameron Sempra (~287) — ~670 MMcf/d combined — are tagged with the gas_day requested by the CLI but actually contain values from the previous gas day. The data is wrong, not missing, which is harder to spot.

Fix: (a) audit which prefixes the portal currently publishes; (b) when no exact match for the requested day is found, do NOT fall back to opts[0] — either raise ParseError (so the day is treated as missing) or restrict the fallback to options whose value contains gas_day_token. The current fallback is silently wrong.

3.2 Part-5 prompt is internally inconsistent

[CRITICAL] src/ng_feedgas/analysis/prompt.py + analysis/stats.py + storage/export.py.

The current 2026-05-28 prompt file shows:

```
Sabine Pass: 2,567 ... (sums to 8,758) TOTAL U.S.: 10,478 MMcf/d
```

Two different "U.S. totals" are computed by different code paths in the same command output. The per-terminal lines come from TERMINAL_ORDER (hardcoded 8 LNG terminals in storage/export.py). The TOTAL U.S. comes from stats.us_total which is sum(terminal_totals()) over the entire DB — i.e. it silently includes Mexico + Canada-Niagara deliveries. The 1,720 MMcf/d delta is hidden.

Fix: make TERMINAL_ORDER config-derived (this is masterplan item #1), and have stats.us_total sum only the displayed terminals (or display every category the way the dashboard does). The masterplan flags this as "highest-value low-risk" work — it is more urgent than that, because the prompt is the headline analyst-facing artifact.

3.3 SQLite races between intraday-fast and intraday-slow

[CRITICAL] data/logs/intraday_fast.log; storage/db.py connect().

Both intraday tasks run on overlapping cadences (5 min and 15 min) and both write to data/feedgas.db. The default SQLite journal mode is rollback (locks the whole DB during a write transaction). Two completed task runs ended with exit=1 today:

```
sqlite3.OperationalError: database is locked ===== 2026-05-28 11:02:49 intraday FAST
END (exit=1) =====
```

Fix: in storage/db.py connect(), enable WAL mode and a busy_timeout on the connection:

```
conn.execute('PRAGMA journal_mode=WAL') conn.execute('PRAGMA busy_timeout=5000')
```

Also reconsider whether running fast/slow tasks at non-coprime intervals (5 / 15) is wise — every 15 minutes they collide by design.

3.4 Cameron / Calcasieu Pass AIS bounding boxes overlap

[HIGH] src/ng_feedgas/calibration/ais.py lines 60-69, _which_terminal().

The two LNG facilities sit roughly 1.5 nm apart on the Calcasieu Ship Channel. The configured boxes overlap in the entire band 29.76-29.81 lat, 93.27-93.36 lon. _which_terminal() iterates LNG_TERMINAL_BOXES dict order and returns the first match — Cameron LNG, since it appears earlier in the dict than Calcasieu Pass. Result: every vessel actually berthed at Calcasieu Pass is logged as "at Cameron LNG". The 394 Cameron observations vs 0 Calcasieu observations in the DB are not coincidence — they are at minimum partly contaminated.

Fix: tighten each box to its actual berth coordinates (Cameron LNG approx 29.794, -93.331; Calcasieu Pass approx 29.781, -93.343) and shrink to ~1 km square. Alternatively, switch _which_terminal() to a "closest centroid" rule.

3.5 Plaquemines nameplate stale — every-pull validator noise

[HIGH] src/ng_feedgas/config/meter_points.yaml line 155.

Plaquemines is configured with nameplate_mmcfd: 2000. Venture Global has begun Phase 2 commissioning and is reportedly already running ~4 Bcf/d feedgas (received authorization for 3.85 Bcf/d). Captured flow (2,491) exceeds the configured nameplate by ~25%, so the validator prints "Plaquemines: total 2,491 MMcf/d is >10% above nameplate 2,000" on every single pull. Fix: bump nameplate to 3,600 MMcf/d (Phase 1+2 design). Calcasieu Pass nameplate (1,500) should also be 1,600 (peak design).

3.6 Iroquois — direction-column logic is dead, docstring is stale

[MEDIUM] src/ng_feedgas/scrapers/iroquois.py lines 1-30, 286-292.

Iroquois' grid uses "Y"/"N" for the Flow Ind column (active/inactive), not "R"/"D". The _dir() function checks for "receipt"/"r"/"delivery"/"d" — it always returns None for "Y", silently falling back to mp.direction from YAML. Works in practice, but the design intent (read direction from the source) is non-functional. The top-of-file docstring also documents "Loc 90 / Waddington" and "Loc 226 / Brookfield" while the YAML uses meter_ids 2250 and 2256.

3.7 Waddington validator warning is structurally misleading

[MEDIUM] storage/db.py terminal_totals() + validators.py.

terminal_totals() filters direction='delivery' only. Waddington's real flow (124 MMcf/d) is direction='receipt' (it's a Canada-to-US import). So the validator sees only Brookfield's 7.5 MMcf/d delivery side and warns "Waddington 8 MMcf/d <20% of nameplate" every run. The dashboard handles imports correctly via _is_canada_import — but the CLI validator does not. Fix: extend terminal_totals() (or add a parallel direction-aware variant) to handle import terminals.

3.8 Cameron LNG via Columbia Gulf shows 0.0 MMcf/d

[MEDIUM] scrapers/tceconnects.py.

Every day, tceconnects matches loc=4246 ("Cameron LNG", BD) on CGT but TSQ is 0. Could be real (idle interconnect) or a stale row — verify by downloading the CGT CSV manually for a recent day and checking whether there are multiple rows containing "Cameron LNG" (only the highest-TSQ one should win, which `_row_match` does for name fallback but not for ID-matched hits).

4. Code-quality findings

4.1 Structural

- **[MEDIUM] No tests, no CI, no packaging.** The masterplan acknowledges this. `_match_meters()`, `_parse_number()`, `upsert_flows()`, and the cycle-mapping dicts are all pure functions — easy first targets. A small pytest suite + GitHub Actions would catch the TETCO regression immediately.
- **[MEDIUM] Hardcoded Python interpreter path in PowerShell scripts.** `tools/*.ps1` all reference `C:\Users\Intern\AppData\Local\Programs\Python\Python312\python.exe`. Breaks on any other workstation.
- **[MEDIUM] `datetime.utcnow()` is deprecated in Python 3.12.** Used throughout scrapers + `ais.py` + db helpers. Switch to `datetime.now(timezone.utc)`.
- **[MEDIUM] Retry logic only exists in KMI.** Williams, Enbridge, ET-iPost, Williams-NWP, tceconnects, iroquois have no retries — a transient HTTP 5xx silently drops the day's data for that scraper. Lift KMI's `_download_with_retry` into `base.py`.
- **[LOW] TGP is downloaded twice per run.** The "kmi" scraper has TGP meters (Niagara) and "tcenergy" delegates to KMIScraper with its own TGP meters (Cameron, Plaquemines). Both groups separately POST the TGP `xlsx`.
- **[LOW] Five scrapers ignore `--date`.** `energytransfer`, `et_ipost`, `williams_nwp`, `tceconnects`, `iroquois` all return the most-recent posted snapshot regardless of the `gas_day` argument. Backfill is impossible for these. Worth documenting explicitly in the CLI help text.

4.2 Data integrity

- **[HIGH] No verification that scraper-stored `gas_day` matches the data's reported `gas_day`.** Most portals echo a "selected gas day" header back into the page/CSV. The scrapers parse only the rows. Capturing the portal's stated gas day and asserting it matches `ctx.gas_day` would have caught the TETCO bug instantly.
- **[MEDIUM] AIS LNG_TANKER_TYPE_RANGE = (80, 89) is too broad.** AIS type 80-89 covers all tankers (oil, chemicals, products, LNG). The 200m-length fallback partly mitigates, but a moored oil tanker at an LNG terminal would still trigger a "60% nameplate" estimate. Tighten to type 80 + 84 + min length 250m.
- **[MEDIUM] "60% of nameplate when ship at berth" is unvalidated.** The README admits this is a heuristic. There is enough EIA monthly data in the DB to calibrate it. Once 30+ days of overlap exist, compute the empirical load factor per terminal and store it.
- **[MEDIUM] Dashboard `signed_mmcfd()` is dead code.** Function in `dashboard.py:206` returns abs values for both directions and is never used in a way that requires signing. KPI label "Canada border (net)" is misleading — Sumas (1,012 receipt) and Niagara (578 delivery) are summed positively into one number.

4.3 Documentation

- **[HIGH] README.md is severely stale.** Describes a 6-of-8-terminal POC at 6,911 MMcf/d. The system now tracks 17 terminals + Mexico + Canada at 10,365 MMcf/d. MASTERPLAN.md is current and correct; README should be replaced or merged.
- **[MEDIUM] docs/portal_notes.md is stale (acknowledged in MASTERPLAN section 8).**
- **[LOW] iroquois.py top-of-file docstring mismatches the YAML.** "Loc 90 / Waddington — RECEIPT, Loc 226 / Brookfield — DELIVERY" but YAML uses meter_ids 2250 and 2256. Both right and wrong in the same file.

5. Recommended improvements

5.1 Immediate (this week)

- Fix the TETCO date fallback (raise `ParseError` instead of silently picking newest `TIMELY`). One line of logic; the highest-impact accuracy improvement available.
- Enable WAL mode + `busy_timeout` on the SQLite connection in `storage/db.py`. Two lines, eliminates the race.
- Unify the Part-5 prompt: either include Mexico + Canada line items, or restrict TOTAL U.S. to the LNG terminals shown. Either is fine; the current mismatch is not.
- Bump Plaquemines nameplate to 3,600 (Phase 2 design). Calcasieu Pass to 1,600.
- Tighten Cameron / Calcasieu Pass AIS bounding boxes so they don't overlap.

5.2 Short term (next 2-4 weeks)

- Make `storage/export.py` `TERMINAL_ORDER` config-derived (masterplan #1). Removes the silent-drop bug for Mexico + Canada in the CLI report.
- Add a 10-line `pytest` suite around `_match_meters` / `_parse_number` / `upsert_flows` / `cycle_dicts`, and a GitHub Actions workflow.
- Lift KMI's `retry` logic into `BaseScraper` so every scraper benefits.
- Add a "source-day verification" step in each scraper: after parsing, capture the portal's reported gas day from the page header, compare to `ctx.gas_day`, raise `ParseError` on mismatch.
- Migrate `datetime.utcnow()` to `datetime.now(timezone.utc)` project-wide.
- Add a "Calcasieu Pass" proxy: even if Venture Global TransCameron is private, EIA monthly + AIS-at-berth could populate a daily proxy.

5.3 Medium term

- Write Golden Pass scraper (gasnom.com ColdFusion). ~2.5 Bcf/d gap — the single biggest remaining miss after the private feeders.
- Write Empire / Chippawa (National Fuel Gas) scraper for the remaining Canada border crossing.
- Build the CENAGAS PDF parser (skeleton in `tools/cenagas_pdf_backfill.py`). The only public source for Texas-intrastate Mexico crossings (Trans-Pecos, Comanche Trail, NET Mexico, Valley Crossing).
- Add a "scraped vs EIA" reconciliation chart: monthly captured-LNG / EIA monthly LNG = capture ratio, plotted over time. Today you can only eyeball it.

- Move the AIS 60% load-factor assumption to a config table that gets calibrated against EIA monthly data.

5.4 Higher-level design

- Add a "data quality" column to every FlowRecord: `actual_gas_day_from_portal`, `scrape_attempt_n`, `parse_warnings`. Surfaces silent-data-bugs in the UI.
- Record the portal's posted-at time when available. This makes "intraday flow change" measurable instead of confounded by scrape latency.
- Consider Postgres instead of SQLite as data volume grows. Today (91 rows) is fine; once you have a year of multi-cycle data (~50k rows) and concurrent intraday writes, Postgres + WAL is more comfortable.
- Add a `/healthz` endpoint to `dashboard.py` that reports last-pull age per scraper.

6. Sources used for accuracy cross-check

Industry / EIA references used to validate the scraped numbers:

- EIA STEO May 2026 — U.S. LNG exports 17.6 Bcf/d (April 2026), 18.1 Bcf/d (March 2026).
- Pipeline & Gas Journal — "Golden Pass Return Fails to Lift U.S. LNG Feedgas Flows" (May 2026). Sabine Pass 3.4 Bcf/d 16-week low on 2026-05-19.
- East Daley — "Plaquemines Demand Passes 2.6 Bcf/d as Venture Global Starts Phase 2".
- S&P Global Commodity Insights — Calcasieu Pass 1.3 baseload / 1.6 peak Bcf/d.
- RBN Energy — Cove Point feedgas 0.75 Bcf/d nameplate.
- Northeast Gas Association — TGP Niagara Expansion 158 MMDth/d.
- Iroquois Gas Transmission company materials — Waddington up to 1.2 Bcf/d receipt capacity.
- TC Energy — North Baja Pipeline 500 MMcf/d southbound design capacity.
- Williams 2026 Q1 8-K — Northwest Pipeline expansions (Silver Spur, Naughton, Wild Trail).

Appendix — how the verification was done

A. Read every Python module under `src/ng_feedgas/` and `dashboard.py` for control flow, error handling, and unit assumptions.

B. Queried `data/feedgas.db` directly to compare what the code claims to write against what is actually stored, per gas-day and per cycle.

C. Replayed the latest week of `intraday_fast.log`, `intraday_slow.log`, `daily_pull.log`, and `ais_track.log` to find runtime errors, warnings, and silent fallbacks.

D. Ran `python -m ng_feedgas show` and `python -m ng_feedgas report` from this environment to inspect actual CLI outputs (CSV + Part-5 prompt).

E. Web-searched EIA, S&P Global, RBN, Pipeline & Gas Journal, NGI, East Daley, and Venture Global filings to anchor expected per-terminal feedgas figures for May 2026.